

Q. Explain the process of creating a database and data dictionary.

To create a new Oracle database and make it available for use, follow these steps as and followed in ocdes:

1. Backup existing databases:

Purpose:

- o To prevent accidental data loss or corruption of existing database during the creation of a new one.

What to backup:

- o Data files
- o Redo log files
- o Control files

⇒ Perform a complete backup using Oracle's backup tools or other reliable methods.

2. Create parameter files:

Purpose: Parameter files contain configuration settings needed to start the database instance.

Steps:

- o Use OS file copy command to duplicate the default parameters

file provided by the Oracle.

- o Rename the copied file to something unique for the new database.

- o Each database should have its own parameter file to avoid conflicts.

3. Edit new parameter files:

⇒ Edit the following parameters:

DB_NAME:

- o Name of the database.

DB_DOMAIN:

- o Network domain of the db.

CONTROL FILES:

- o Paths to control files.

DB_BLOCK_SIZE:

- o Size of the db blocks.

DB_BLOCK_BUFFERS:

- o Number of db buffers.

PROCESSES:

- o Number of processes allowed.

ROLLBACK_SEGMENTS:

- o Rollback segment information.

License parameters

LICENSE MAX_SESSIONS:

- o maximum sessions allowed.

LICENSE SESSION_WARNING:

- o session warning limit

LICENSE MAX_USERS:

- o maximum users allowed.

4. Check the instance identifier:

Purpose:

Ensure that the Oracle instance identifier matches the db name to avoid confusion with other db.

How to check:

- o Look at the current instance identifier settings on system.
- o verify that it matches the DB_NAME parameter set in the new parameters file.

5. Start SQL *PLUS and connect as SYSDBA:

Purpose: SQL *Plus is used to interact with the Oracle database.

- o SYSDBA privileges are required to perform administrative tasks.

Commands:

- o Open SQL*PLUS without logging in:
- o SQLPLUS /nolog
- o Connect as SYSDBA: connect username / password as sysdba

6. Start an instance:

Purpose:

Start the Oracle instance (SGA and background processes) in preparation for creating the db.

Command:

- o STARTUP NOMOUNT;

Explanation:

- o This command starts the instance without mounting the db, meaning the SGA is allocated, and background processes start running, but the db itself is not yet available.

Create the DB:

Purpose

- Create the new database and its associated files.

Command

- Use the CREATE DATABASE SQL statement

```
CREATE DATABASE new_database_name
USER SYS IDENTIFIED BY password
USER SYSTEM IDENTIFIED BY password
LOGFILE GROUP 1 ('/path/to/redo01.log')
            SIZE 50M,
        GROUP 2 ('/path/to/redo02.log') SIZE 50M
MAXLOGFILES 5
DATAFILES ('Path'...) SIZE 500M
MAXDATAFILES 100
CHARACTER SET AL32UTF8
NATIONAL CHARACTER SET AL16UTF16
EXTENT MANAGEMENT LOCAL
DEFAULT TEMPORARY TABLESPACE TO
TEMPFILE 'Path/to/temp01.dbf' SIZE 50M
UNDO TABLESPACE undo DATAFILE 'Path'
SIZE 200M;
```

Explanation:

- Create datafiles
- Create control files
- Create redo log files
- Creates the SYSTEM tablespace and
- Rollback segment
- Creates the data dictionary
- Creates users SYS and SYSTEM
- Specifies the character set
- Mounts and opens the database

Final step: Backup the database:

Purpose: To ensure that a backup of the newly created database.

Recommendation:

- Immediately perform backup after creation, including all critical files (parameter files, data files, redo log files, control files).

Note: Creating an Oracle database requires planning and is done in multiple steps.

⇒ You create db only once, but can change the configuration (except the block size) or add more files to db.

⇒ Before creating db you must have the following:

⇒ Necessary hardware resources such as memory and disk space.

⇒ OS and SYSDBA privileges

⇒ A plan to lay out the files and their sizes

⇒ A parameter file

⇒ Defined environmental variables

⇒ Oracle software installed

⇒ A backup of existing db

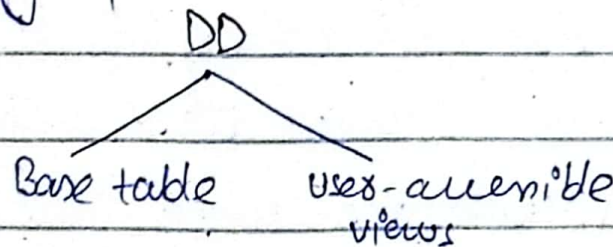
⇒ Database can be created in two ways:

- manually

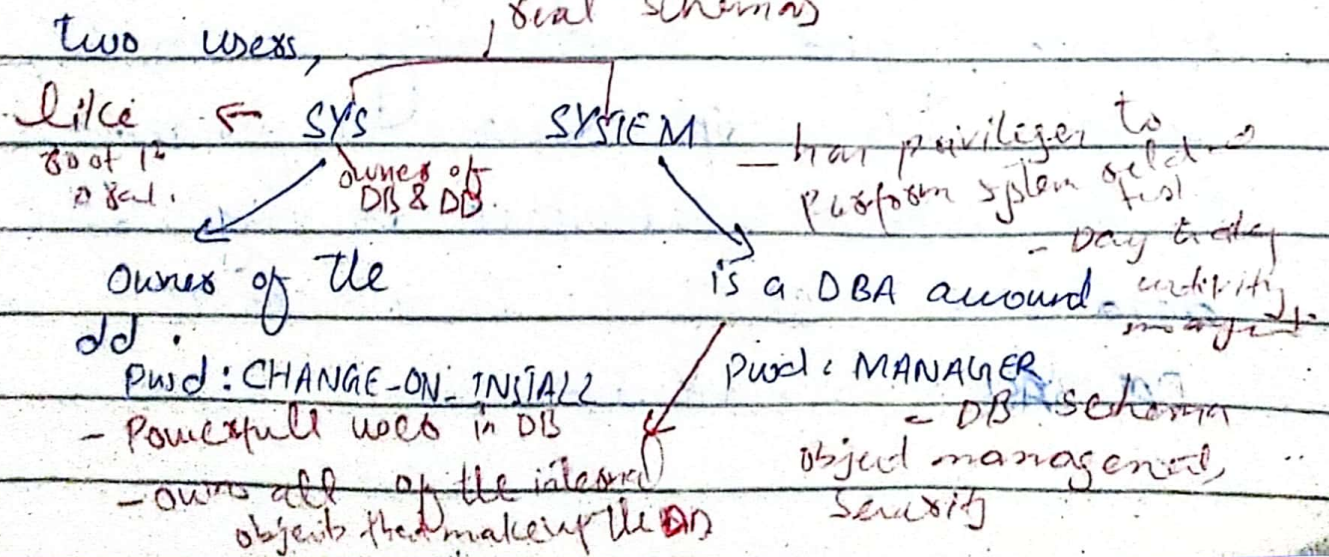
- Database Configuration Assistant

Process of Creating DB dictionary

- o The important part of the Oracle db is the data dictionary.
- o The data dictionary is a set of tables and views that holds database metadata information.
- o Cannot update the dictionary directly. Oracle updates the dictionary when issue any Data-definition Language commands.
- o The dictionary is provided as read only for users and administrators.



when the db is created, Oracle creates two users, *system schemas*



⇒ Oracle uses the dictionary information for proper functioning of the db.

⇒ The Oracle db is functional only when create the dictionary views and additional tablespace, rollback segments, users and so on.

⇒ Creating the dictionary views in the next step after create the database by using CREATE DATABASE command.

- o Running certain Oracle-supplied scripts creates the dictionary views.

Creating dictionary scripts

⇒ The data dictionary base tables are created under the SYS schema in the SYSTEM Tablespace when issue the CREATE DATABASE command.

⇒ Oracle automatically creates the tablespace and tables using the sql.bsr script. This script creates the following:

- o The SYSTEM tablespace by using the data files specified in the CREATE DATABASE command.

- o A rollback segment named SYSTEM in the SYSTEM tablespace.

- o The SYS and SYSTEM users accounts.
- o The dictionary base tables and clusters.
- o Indexes on dictionary tables and sequences for dictionary use.
- o The DUAL table.

The roles PUBLIC, CONNECT, RESOURCE, DBA, DELETE_CATALOG_ROLE, EXECUTE_CATALOG_ROLE, and SELECT_CATALOG_ROLE.

Creating views

- o Running the script catalog.sql creates the data dictionary views.

This script creates synonyms on the views to allow users easy access to the views.

Before running any data dictionary script, connect to the db as SYS.

Creating dictionary items

- o The script catproc.sql creates the dictionary items necessary for PL/SQL functionality.

Creating dictionary objects

- Scripts necessary for creating dictionary objects depend on the OS and the functionality that we want in the database.

- At a minimum, run the catalog.sql and catproc.sql scripts after creating the database.

⇒ The dictionary creation scripts all begin with cat. many of the scripts call other scripts. for example, when execute catalog.sql, it calls the following scripts:

- standard.sql

↳ Creates a package called STANDARD.

- cataudit.sql

↳ Creates data dictionary views to support auditing.

- catexp.sql

↳ to support import/export.

- catdx.sql

To support direct path load of SQL * Loader

o catpart.sql

- Support partitioning

o catadt.sql

- Support oracle objects and types.

o catsum.sql

- Support oracle summary management

Note:

cat*.sql : catalog and data,
dictionary scripts.
